

**AIRBUS**

NORMAL CHECKLIST

A330-200/300

COCKPIT PREPARATION

GEAR PINS and COVERS.....REMOVED
FUEL QUANTITY.....KG/LB
SIGNS.....ON/AUTO
ADIRS.....NAV
BARO REF.....(BOTH)

BEFORE START

PARKING BRAKE.....
T.O SPEEDS & THRUST.....(BOTH)
WINDOWS/DOORS.....CLOSED (BOTH)
BEACON.....ON

AFTER START

ANTI-ICE.....AS RQRD
ECAM STATUS.....CHECKED
PITCH TRIM.....% SET
RUDDER TRIM.....NEUTRAL

OETD

AFTER ENG 1 START:
X-BLEED.....OPEN
APU BLEED.....OFF
APU.....KEEP ON
AFTER START PROC.....APPLY

Delay ECAM STATUS check and the wing anti-ice setting until all engines are started.

ENG 2 START minimum 3 min (5 min if engine cold) before T/O (PM):
APU BLEED.....ON
ENG MODE SEL.....IGN/START
ENG 2 MASTER (after 10s).....START
APU.....AS RQRD
X-BLEED.....AUTO
AFTER START PROC.....APPLY

TAXI

FLT CTL.....CHECKED (BOTH)
FLAP SETTING.....CONF ____ (BOTH)
ENG MODE SEL.....
RADAR & PRED W/S.....ON
ECAM MEMO.....TO NO BLUE
- SEAT BELTS ON
- CABIN CHECK
- SPLRS ARM
- FLAPS T.O
- AUTO BRK MAX
- T.O CONFIG NORMAL

LINE-UP

T.O RWY.....(BOTH)
CABIN CREW.....ADVISED
TCAS.....TA OR TA/RA
PACK 1 & 2.....(BOTH)

DEPARTURE CHANGE

RWY & SID.....CONFIRM
FLAP SETTING.....CONF ____ (BOTH)
T.O SPEEDS & THRUST.....(BOTH)
FCU ALT.....

APPROACH

BARO REF.....(BOTH)
SEAT BELTS.....ON
MINIMUM.....(BOTH)
AUTO BRAKE.....AS RQRD
ENG MODE SEL.....

LANDING

G/A ALTITUDE.....SET (BOTH)
CABIN CREW.....ADVISED
ECAM MEMO.....LDG NO BLUE
- SEAT BELTS ON
- CABIN CHECK
- LDG GEAR DN
- FLAPS LDG
- SPLRS ARM

AFTER LANDING

RADAR & PRED W/S.....OFF

OETA

APU.....START
Not less than 2 minutes after landing and when taxiing in a straight line:
ENG 2.....SHUT DOWN
X-BLEED.....OPEN

PARKING

APU BLEED.....AS RQRD
ENGINES.....OFF
SIGNS.....OFF
EXT LT.....AS RQRD
FUEL PUMPS.....OFF
PARK BRK and CHOCKS.....AS RQRD
TRANSPONDER.....STANDBY 2000

SECURING THE AIRCRAFT

ADIRS.....OFF
OXYGEN.....OFF
APU BLEED.....OFF
EMER EXIT LT.....OFF
SIGNS.....OFF
EXT PWR.....AS RQRD
APU AND BAT.....OFF

Consider COLD WEATHER

EMERGENCY EVACUATION

AIRCRAFT/PARKING BRAKE.....STOP / ON
 ATC (VHF 1).....NOTIFY
 CABIN CREW (PA).....ALERT
 ΔP (Only is MAN CAB PR has been used).....CHECK ZERO

● **If ΔP not at zero:**

CAB PR MODE SEL.....MAN
 V/S CTL.....FULL UP

ALL ENG MASTERS.....OFF
 ALL FIRE pb (ENGs & APU).....PUSH
 ALL AGENTS (ENGs & APU).....AS RQRD

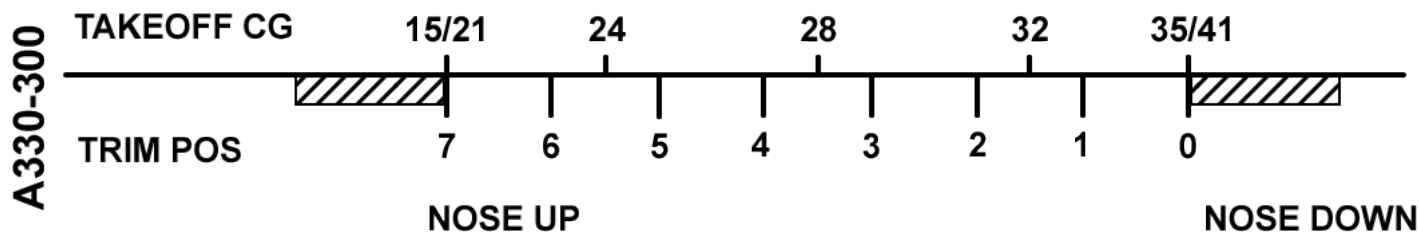
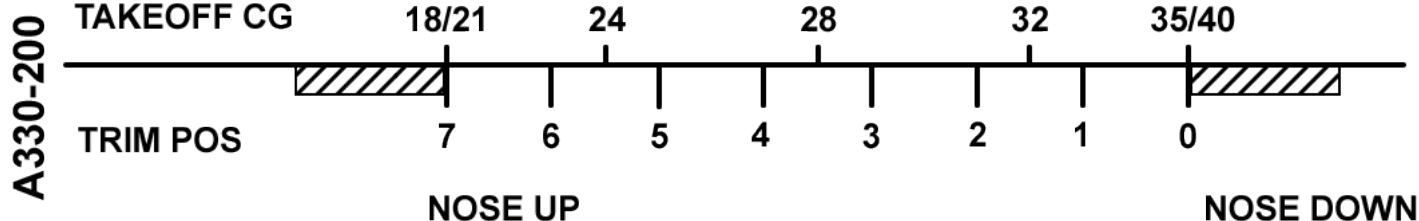
■ **If evacuation required:**

EVACUATION.....INITIATE
 ALL BATs.....OFF

■ **If evacuation NOT required:**

CABIN CREW AND PASSENGERS (PA).....NOTIFY

TAKEOFF CG/TRIM





ENGINE START PROCEDURE

A330-200/300

AUTOMATIC ENGINE START

THRUST levers.....IDLE

ENG START selector.....IGN/START
The lower ECAM display shows ENG page

ANNOUNCE.....“STARTING ENGINE 1”

At least 3 s after the ENG START selector is set to the IGN START position and when all amber crosses and messages have disappeared on the engine parameters (upper ECAM display)

ENG MASTER 1.....ON

■ If GE Engines Installed:

ON ECAM UPPER DISPLAY	ON ECAM LOWER DISPLAY
N2 increases	Corresponding start valve in line Bleed pressure indication green Oil pressure increases
At 10 % N2	Indication of active igniter (A or B)
At 15 % N2: – FF increases 20 seconds (maximum) after fuel is on: – EGT increases – N1 increases	
At 50 % N2	Start valve cross line
At 54 % N2	Igniter indication off

MAIN AND SECONDARY ENG. IDLE PARAMETERS.....CHECK NORMAL
*At ISA sea level:
N1 about 23 %
N2 about 63 %
EGT about 360 °C
FF about 550 kg/h (1 210 lb/h)*

ANNOUNCE.....“STARTING ENGINE 2”

ENG MASTER 2.....ON
Same procedure as for engine 1

■ If RR Engines Installed:

ON ECAM UPPER DISPLAY	ON ECAM LOWER DISPLAY
N3 increases	Corresponding start valve in line Bleed pressure indication green Oil pressure increases
At 25 % N3: – FF increases 30 seconds (maximum) after fuel is on: – EGT increases – N1 increases	Indication of active igniter (A or B)
At 50 % N3:	Start valve starts closing (It is fully closed after about 10 s).

MAIN AND SECONDARY ENG. IDLE PARAMETERS.....**CHECK NORMAL**
At ISA sea level:
EPR about 1.015
N1 about 22.6 %
N2 47 %
N3 about 63 %
EGT 380 °C
FF 820 kg/h (1 800 lb/h)

ANNOUNCE.....**“STARTING ENGINE 2”**
 ENG MASTER 1.....**ON**
Same procedure as for engine 1

■ If PW Engines Installed:

ON ECAM UPPER DISPLAY	ON ECAM LOWER DISPLAY
N2 increases	Corresponding start valve in line Bleed pressure indication green Oil pressure increases
At 20 % N2: – FF increases 30 seconds (maximum) after fuel is on: – EGT increases – N1 increases	Indication of active igniter (A or B)
Between 43 % and 48 % N2	Start valve closes Igniter indication off

MAIN AND SECONDARY ENG. IDLE PARAMETERS.....**CHECK NORMAL**
At ISA sea level:
EPR about 1.010
N1 about 23 %
N2 about 59 %
EGT 320 °C
FF 580 kg/h (1 280 lb/h)

ANNOUNCE.....**“STARTING ENGINE 2”**
 ENG MASTER 1.....**ON**
Same procedure as for engine 1



MANUAL ENGINE START

THR LEVERS.....IDLE

ENG MODE selector.....NORM THEN IGN/START
The lower ECAM displays engine page

NOTE: *If both engines are started manually, the following procedure applies one engine at a time.*

■ If GE engines installed:

- **When all engines parameters are available on the upper ECAM display (no amber crosses displayed):**
ENG MAN START pb-sw.....ON
START VALVE.....CHECK IN-LINE
N2 INCREASE.....CHECK
OIL PRESS INCREASE.....CHECK
- **When N2 reaches the maximum motoring speed (15 % minimum):**
ENG MASTER.....ON
CHRONO.....START
The PM starts the timing in order to monitor the light-up duration
IGNITERS A AND B.....CHECK ON
FUEL FLOW INCREASE.....CHECK
- **20 s maximum after fuel flow increase:**
EGT INCREASE.....CHECK
N1 INCREASE.....CHECK
- **When N2 reaches 50 %:**
START VALVE (is fully closed after 3 to 4 s).....CHECK CROSS LINE
IGNITERS A AND B (at or above 54 % N2).....CHECK OFF
MAIN ENG PARAMETERS.....CHECK NORMAL
SECONDARY ENG PARAMETERS.....CHECK NORMAL
- **When idle is reached (AVAIL indication is displayed):**
ENG MAN START pb-sw.....NORM
ENG START selector.....NORM
SOP – ENGINE START.....RESUME



■ **If RR engines installed:**

- **When all engines parameters are available on the upper ECAM display (no amber crosses displayed):**
ENG MAN START pb-sw.....ON
START VALVE.....CHECK IN-LINE
N3 INCREASE.....CHECK
OIL PRESS INCREASE.....CHECK
- **When N3 reaches the maximum motoring speed (25 % minimum):**
ENG MASTER.....ON
CHRONO.....START
The PM starts the timing in order to monitor the light-up duration
IGNITERS A AND B.....CHECK ON
FUEL FLOW INCREASE.....CHECK
- **30 s maximum after fuel flow increase:**
EGT INCREASE.....CHECK
N1 INCREASE.....CHECK
- **When N3 at or above 50 %:**
START VALVE.....CHECK CROSS LINE
IGNITERS A AND B.....CHECK OFF
- **When idle is reached (AVAIL indication is displayed):**
MAIN ENG PARAMETERS.....CHECK NORMAL
SECONDARY ENG PARAMETERS.....CHECK NORMAL
ENG MAN START pb-sw.....NORM
ENG START selector.....NORM
SOP – ENGINE START.....RESUME



■ If PW engines installed:

- **When all engines parameters are available on the upper ECAM display (no amber crosses displayed):**
ENG MAN START pb-sw.....ON
START VALVE.....CHECK IN-LINE
N2 INCREASE.....CHECK
OIL PRESS INCREASE.....CHECK
- **When N2 reaches the maximum motoring speed (15 % minimum):**
ENG MASTER.....ON
CHRONO.....START
The PM starts the timing in order to monitor the light-up duration
IGNITERS A AND B.....CHECK ON
FUEL FLOW INCREASE.....CHECK
- **30 s maximum after fuel flow increase:**
EGT INCREASE.....CHECK
N1 INCREASE.....CHECK
- **When N2 between 43 % and 48 %:**
START VALVE.....CHECK CROSS LINE
IGNITERS A AND B.....CHECK OFF
- **When idle is reached (AVAIL indication is displayed):**
MAIN ENG PARAMETERS.....CHECK NORMAL
SECONDARY ENG PARAMETERS.....CHECK NORMAL
ENG MAN START pb-sw.....NORM
ENG START selector.....NORM
SOP – ENGINE START.....RESUME



MEMORY ITEMS

A330-200/300

LOSS OF BRAKING

- **If no braking:**

REV.....MAX

If needed, the flight crew may keep maximum reverse thrust until full aircraft stop

BRAKE PEDALS.....RELEASE

A/SKID & N/W STRG.....OFF

BRAKE PEDALS.....PRESS

MAX BRK PR.....1000 PSI

Limit brake pressure to approximately 1 000 PSI. Monitor brake pressure on the BRAKES PRESS indicator

- **If still no braking:**

PARK BRAKE.....USE

Use short successive parking brake applications to stop the aircraft.

EMER DESCENT

CREW OXY MASKS.....USE

SIGNS.....ON

EMER DESCENT.....INITIATE

- **If A/THR not active:**

THR LEVERS.....IDLE

SPEED BRAKES.....FULL

- **When descent established:**

SPEED.....MAX / APPROPRIATE

- **If structural damage suspected:**

MANEUVER WITH CARE

CONSIDER L/G EXTENSION

ENG MODE SEL.....IGN

ATC.....NOTIFY

EMER DESCENT (PA).....ANNOUNCE

ATC XPDR 7700.....CONSIDER

MAX FL.....100/MEA-MORA

- **If CAB ALT above 14 000 ft:**

OXYGEN PAX MASK MAN ON.....PRESS



MEMORY ITEMS

A330-200/300

STALL RECOVERY

NOSE DOWN PITCH CONTROL.....**APPLY**
In case of lack of pitch down authority, reducing thrust may be necessary

BANK.....**WINGS LEVEL**

- **When out of stall (no longer stall indications):**

THRUST.....**INCREASE SMOOTHLY AS NEEDED**

SPEED BRAKES.....**CHECK RETRACTED**

FLIGHT PATH.....**RECOVER SMOOTHLY**

- **If in clean configuration and below 20 000 ft:**

FLAP 1.....**SELECT**

STALL WARNING AT LIFTOFF

THRUST.....**TOGA**

At the same time:

PITCH ATTITUDE.....**15°**

BANK.....**WINGS LEVEL**

GPWS/EGPWS Warnings

- **“PULL UP” – “TERRAIN AHEAD PULL UP” – “OBSTACLE AHEAD PULL UP” – “AVOID TERRAIN” – “AVOID OBSTACLE”**

Simultaneously:

AP.....**OFF**

PITCH.....**PULL UP**

Pull to full backstick and maintain in that position

THRUST LEVERS.....**TOGA**

SPEED BRAKES lever.....**CHECK RETRACTED**

BANK.....**WINGS LEVEL or ADJUST**

DO NOT CHANGE CONFIGURATION (SLATS/FLAPS, GEAR) UNTIL CLEAR OF OBSTACLE.

TCAS WARNING – RESOLUTION ADVISORY

Always follow RA orders, even if this results in crossing the intruder altitude, because these orders ensure the best altitude separation.

CAUTION Be aware that the intruder may have a TCAS, and may maneuver in response to a coordinated RA order. Therefore, not following an RA order could compromise safe separation.

■ If the AP/FD **TCAS** mode is available:

The flight crew applies this procedure, when a RA triggered, and the AP/FD **TCAS** mode is engaged. The AP/FD **TCAS** mode follows the RA orders.

● All RA, except any **CLIMB** RA during approach in **CONF 3** or **FULL**:

● If the **AP** is **OFF**:

FD ORDERs.....**FOLLOW**

The AP can be engaged

VERTICAL SPEED.....**MONITOR**

If a preventative RA was triggered: Check that the vertical speed remains out of the red areas of the vertical speed scale.

If a corrective RA was triggered: Check that the vertical speed remains out of the red area, and remains in the green area of the vertical speed scale.

CAUTION If for any reason during a RA, the aircraft vertical speed does not reach the green area of the vertical speed scale, the Pilot Flying (PF) should disconnect the AP, and override the FD orders, in order to lead the aircraft vertical speed out of the red area of the vertical speed scale. If necessary, the PF must use the full speed range between Vmax and Vmax.

● Any **CLIMB** RA during approach in **CONF 3** or **FULL**:

GO-AROUND.....**PERFORM**

The AP/FD **TCAS** mode disengages (the AP/FD no longer follows the RA orders).
Follow the **SRS** GA mode.

VERTICAL SPEED.....**MONITOR**

Check that the vertical speed remains out of the red area of the vertical speed scale and take over if necessary.

Respect stall, GPWS or windshear warnings.

ATC.....**NOTIFY**

- When the “CLEAR OF CONFLICT” aural alert sounds:

The AP/FD TCAS mode, if engaged, disengages

AP/FD.....	MONITOR/FOLLOW
ATC.....	NOTIFY
LATERAL AND VERTICAL GUIDANCE.....	ADJUST

The flight crew should engage and appropriate vertical mode, or adjust the vertical speed target, in accordance with the latest ATC clearance.

SPEED.....	ADJUST
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The flight crew should adjust the speed target, and revert to managed speed, as appropriate.

- If the AP/FD TCAS mode is not available:

The flight crew must apply this procedure, when a RA is triggered and the AP/FD TCAS mode does not engage.

- All RA, except CLIMB RA during approach in CONF 3 or FULL:

AP (if engaged).....	OFF
BOTH FDs.....	OFF

Respond promptly and smoothly.

VERTICAL SPEED.....	ADJUST or MAINTAIN
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Adjusting or maintain the vertical speed as required, to reach the green area and/or avoid the red area of the vertical speed scale.

Note: Avoid excessive maneuvers while attempting to maintain the vertical speed just outside the red area of the vertical speed scale, and within the green area. If necessary, use the full speed range between Vmax and Vmax.

- Any CLIMB RA during approach in CONF 3 or FULL:

GO-AROUND.....	PERFORM
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Follow the SRS GA mode.

VERTICAL SPEED.....	MONITOR
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Check that the vertical speed remains out of the red area of the vertical speed scale and take over if necessary.

Respect stall, GPWS or windshear warnings.

ATC.....	NOTIFY
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**● When the “CLEAR OF CONFLICT” aural alert sounds:**

ATC.....	NOTIFY
LATERAL AND VERTICAL GUIDANCE.....	ADJUST

Adjust the lateral and vertical guidance to resume normal navigation, in accordance with ATC clearance.

AP/FD.....	AS RQRD
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If necessary, reengage the AP/FD.

WINDSHEAR WARNING – REACTIVE WINDSHEAR

If windshear is detected, both PFDs display a **WINDSHEAR** message. An associated aural WINDSHEAR alert sounds three times.

If the flight crew detects WINDSHEAR, apply the following recovery technique:

■ Takeoff:**■ If before V1:**

If there are significant variations in airspeed, and in airspeed trend below the indicated V_1 , reject the takeoff.

If windshear occurs during the takeoff roll, the aircraft may reach V_1 later (or sooner) than expected. The flight crew should ensure that there is sufficient runway remaining to stop the aircraft, if necessary.

■ If after V1:

THR LEVERS.....	TOGA
REACHING VR.....	ROTATE
SRS ORDERS.....	FOLLOW

If necessary, the flight crew may pull the sidestick fully back.

NOTE: *If the FD bars are not displayed, move toward an initial pitch attitude of 17.5° . Then, if necessary, to prevent a loss in altitude, increase the pitch attitude.*

■ Airborne, initial climb or landing:

THR LEVERS AT TOGA.....	SET OR CONFIRM
AP (if engaged).....	KEEP ON
SRS ORDERS.....	FOLLOW

If necessary the flight crew may pull the sidestick fully back.

NOTE: *1. Autopilot disengages if the angle of attack value goes above alpha prot.
2. If the FD bars are not displayed, move toward an initial pitch attitude of 17.5° . Then, if necessary, to prevent a loss in altitude, increase the pitch attitude.*

DO NOT CHANGE CONFIGURATION (SLATS/FLAPS, GEAR) UNTIL OUT OF WINDSHEAR. CLOSELY MONITOR FLIGHT PATH AND SPEED. RECOVER SMOOTHLY TO NORMAL CLIMB OUT OF WINDSHEAR.